

Let $A = \sum (\underline{x}_\alpha - \bar{\underline{x}}) (\underline{x}_\alpha - \bar{\underline{x}})'$

2.) $\sum_{\alpha=1}^N [(\underline{x}_\alpha - \underline{\mu})' \Sigma^{-1} (\underline{x}_\alpha - \underline{\mu})]$

$\text{tr}(CD) = \text{tr}(DC)$

$= \text{tr} \left(\sum_{\alpha=1}^N (\underline{x}_\alpha - \underline{\mu}) \Sigma^{-1} (\underline{x}_\alpha - \underline{\mu})' \right)$

$= \text{tr} \left(\Sigma^{-1} [A + N(\bar{\underline{x}} - \underline{\mu})(\bar{\underline{x}} - \underline{\mu})'] \right)$

↓ using result 1.

$= \text{tr}(\Sigma^{-1}A) + N \text{tr}[(\bar{\underline{x}} - \underline{\mu})' \Sigma^{-1} (\bar{\underline{x}} - \underline{\mu})]$

3) Lemma: If D is positive definite matrix of order $p \times p$, the maximum of $f(G) = -N \log |G| - \text{tr}(G^{-1}D)$ wrt a pd matrix G exists and occurs at $G = \frac{1}{N} D^{-1}$

4) Cholsky decomposition:

If A is pd, $\exists$ a unique lower triangular matrix T ($t_{ii} > 0$ $i < j$) with positive diagonal elements such that $A = TT'$

Eg: $T = \begin{pmatrix} a & 0 & 0 \\ b & d & 0 \\ c & e & f \end{pmatrix} \quad |T|^2 = \begin{vmatrix} a^2 & 0 & 0 \\ ab+bd & d^2 & 0 \\ ac+be & de+ef & f^2 \end{vmatrix}$

$\log |T|^2 = \log a^2 + \log d^2 + \log f^2$

$TT' = \begin{bmatrix} a & 0 & 0 \\ b & d & 0 \\ c & e & f \end{bmatrix} \begin{bmatrix} a & b & c \\ 0 & d & e \\ 0 & 0 & f \end{bmatrix}$

$\text{tr}(TT') = a^2 + b^2 + c^2 + d^2 + e^2 + f^2$

Pf: (Lemma)

Let $D = EE'$ (D is p.d)

and $E'G^{-1}E = H$

$G^{-1} = (E')^{-1} H (E^{-1})$

$$G = E(H^{-1})E'$$

$$\Rightarrow |G| = |E| |H^{-1}| |E'|$$

$$= |H^{-1}| |EE'|$$

$$= |H^{-1}| |D| = \frac{|D|}{|H|}$$

$$\text{tr}(G^{-1}D) = \text{tr}(G^{-1}EE')$$

$$= \text{tr}(E'G^{-1}E)$$

$$= \text{tr}(H)$$

$$f(G) = -N \log |G| - \text{tr}(G^{-1}D)$$

$$= -N \log \frac{|D|}{|H|} - \text{tr}(H)$$

$$= -N \log |D| + N \log |H| - \text{tr}(H)$$

$$(\text{Let } H = TT')$$

$$= -N \log |D| + \cancel{N \log |D|} + \cancel{N \log |D|} + \cancel{N \log |D|} + N \log |T| - \text{tr}(TT')$$

$$= -N \log |D| + \sum_{i=1}^p N \log t_{ii}^2$$

$$- \sum_{i=1}^p (t_{ii}^2) - \left(\sum_{i < j} \sum_{i < j} t_{ij}^2 \right)$$

$$= -N \log |D| + \sum_{i=1}^p (N \log t_{ii}^2 - t_{ii}^2) - \sum_{i < j} \sum_{i < j} t_{ij}^2$$

$$\frac{\partial f}{\partial t_{ii}^2} = \frac{N}{t_{ii}^2} - 1 = 0 \quad \underline{N = t_{ii}^2}$$

$$\frac{\partial^2 f}{(\partial t_{ii}^2)^2} = -\frac{N}{t_{ii}^4} = -\frac{1}{t_{ii}^2} < 0$$

$$H = NI$$

$$\text{Then } G = EH^{-1}E'$$

$$= \frac{1}{N} EE' = \frac{1}{N} D$$

$$f(G) = -N \log |G| - \text{tr}(G^{-1}D)$$

$$f\left(\frac{1}{N}D\right) = -N \log \left| \frac{D}{N^p} \right| - N \operatorname{tr} D^{-1}D$$

$$= -N \log |D| + pN \log N - Np$$

MLE

$$\underline{x}_1, \underline{x}_2, \dots, \underline{x}_N \sim N_p(\underline{\mu}, \Sigma) \quad N > p$$

$$L = \frac{1}{(2\pi)^{\frac{Np}{2}} |\Sigma|^{\frac{N}{2}}} \exp \left[-\frac{1}{2} \sum_{\alpha=1}^N (\underline{x}_\alpha - \underline{\mu})' \Sigma^{-1} (\underline{x}_\alpha - \underline{\mu}) \right]$$

$$\begin{aligned} \log L &= -\frac{Np}{2} \log(2\pi) - \frac{N}{2} \log |\Sigma| - \frac{1}{2} \sum_{\alpha=1}^N (\underline{x}_\alpha - \underline{\mu})' \Sigma^{-1} (\underline{x}_\alpha - \underline{\mu}) \\ &= \underbrace{-\frac{Np}{2} \log(2\pi)}_{\text{const.}} - \frac{N}{2} \log |\Sigma| \quad \textcircled{I} - \frac{1}{2} \operatorname{tr}(\Sigma^{-1} A) \quad \textcircled{II} \\ &\quad - \frac{N}{2} (\bar{\underline{x}} - \underline{\mu})' \Sigma^{-1} (\bar{\underline{x}} - \underline{\mu}) \quad \textcircled{III} \end{aligned}$$

$$\textcircled{III} \quad \Sigma > 0, \quad \Sigma^{-1} > 0$$

$$\Rightarrow (\bar{\underline{x}} - \underline{\mu})' \Sigma^{-1} (\bar{\underline{x}} - \underline{\mu}) \geq 0$$

$$= 0 \quad \text{iff} \quad \underline{\mu} = \bar{\underline{x}}$$

Using Lemma (3.1)

$$f(\Sigma) = -N \log |\Sigma| - \operatorname{tr} [\Sigma^{-1} A]$$

maxim occurs at

$$G \equiv \Sigma$$

$$A \equiv D$$

$$\hat{\Sigma} = A$$

$$\Rightarrow \hat{\Sigma} = \frac{1}{N} \left[\sum_{\alpha=1}^N (\bar{\underline{x}}_\alpha - \bar{\underline{x}})(\bar{\underline{x}}_\alpha - \bar{\underline{x}})' \right]$$

$$\Rightarrow \hat{\underline{\mu}} = \bar{\underline{x}}$$

$$\Rightarrow E(\hat{\underline{\mu}}) = E(\bar{\underline{x}}) = \begin{pmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_p \end{pmatrix} = \underline{\mu}$$

Hence $\hat{\underline{\mu}}$ is unbiased estimator

Define

$$S = \frac{A}{N-1}$$

$$S = \begin{pmatrix} s_{11} & s_{12} & \dots & s_{1p} \\ s_{21} & s_{22} & \dots & s_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ s_{p1} & s_{p2} & \dots & s_{pp} \end{pmatrix}$$

$$E(S) = \begin{pmatrix} \sigma_1^2 & \sigma_{12} & \dots & \sigma_{1p} \\ \sigma_{21} & \sigma_2^2 & \dots & \sigma_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{p1} & \sigma_{p2} & \dots & \sigma_p^2 \end{pmatrix} = \Sigma$$

→ $\hat{\Sigma}_{MLE}$ is biased estimator

→ $\hat{S}$ is unbiased estimator

Distribution of sample mean vector.

$$\underline{X} \sim N_p(\underline{\mu}, \Sigma)$$

$$L(\underline{x}_1, \underline{x}_2, \dots, \underline{x}_n) = \frac{1}{(2\pi)^{np/2} |\Sigma|^{n/2}} \exp \left[-\frac{1}{2} \sum_{\alpha=1}^n (\underline{x}_\alpha - \underline{\mu})^T \Sigma^{-1} (\underline{x}_\alpha - \underline{\mu}) \right]$$

Apply the transformation

$X^* = XQ$, Q is an orthogonal matrix such that first column has elements $\frac{1}{\sqrt{N}}$ and all others are q_{ij}

$$Q = \left(\begin{array}{c|c} \frac{1}{\sqrt{N}} & q_{ij} \end{array} \right) \quad X = \begin{pmatrix} x_{11} & \dots & x_{1N} \\ \vdots & & \vdots \\ x_{p1} & \dots & x_{pN} \end{pmatrix}$$

$$X^* = XQ = [\sqrt{N} \bar{\underline{X}}, \underline{Y}] \text{ reduced matrix}$$

$$X^* X^{*'} = X Q Q' X' = X X'$$

$$X^* X^{*'} = N \bar{\underline{X}} \bar{\underline{X}}' + \underline{Y} \underline{Y}'$$

$$\Rightarrow \underline{Y} \underline{Y}' = X X' - N \bar{\underline{X}} \bar{\underline{X}}'$$

$$\Rightarrow \sum_{\alpha=1}^n \underline{y}_\alpha \underline{y}_\alpha' = \sum_{\alpha=1}^n (\underline{x}_\alpha - \bar{\underline{X}})(\underline{x}_\alpha - \bar{\underline{X}})' = A = (N-1)S$$

$$\begin{aligned}
 L(\underline{X}_1, \dots, \underline{X}_n) &= K^* \exp \left[-\frac{1}{2} \text{tr} \Sigma^{-1} (\underline{x}_n - \underline{\mu}) (\underline{x}_n - \underline{\mu})' \right] \\
 &= K^* \exp \left[-\frac{1}{2} \text{tr} \left\{ \sum_{n=1}^n (\underline{x}_n - \bar{\underline{x}}) (\underline{x}_n - \bar{\underline{x}})' + N(\bar{\underline{x}} - \underline{\mu})(\bar{\underline{x}} - \underline{\mu})' \right\} \right] \\
 &= K^* \exp \left[-\frac{1}{2} \text{tr} \Sigma^{-1} \{ \underline{Y} \underline{Y}' + N(\bar{\underline{x}} - \underline{\mu})(\bar{\underline{x}} - \underline{\mu})' \} \right] \\
 &= K^* \exp \left[-\frac{1}{2} \text{tr} \Sigma^{-1} \underline{Y} \underline{Y}' \right] \cdot \exp \left[-\frac{1}{2} (\bar{\underline{x}} - \underline{\mu})' \left(\frac{\Sigma}{N} \right)^{-1} (\bar{\underline{x}} - \underline{\mu}) \right] \\
 &= \frac{1}{(2\pi)^{\frac{np}{2}} \cdot |\Sigma|^{\frac{n}{2}}} \exp \left[-\frac{1}{2} \underline{Y}' \Sigma^{-1} \underline{Y} \right] \cdot \exp \left[-\frac{1}{2} (\bar{\underline{x}} - \underline{\mu})' \left(\frac{\Sigma}{N} \right)^{-1} (\bar{\underline{x}} - \underline{\mu}) \right]
 \end{aligned}$$

① $\Rightarrow \bar{\underline{X}}$ and $\underline{Y}$ are independently distributed.

② $\Rightarrow \bar{\underline{X}} \sim N\left(\underline{\mu}, \frac{\Sigma}{N}\right)$

③ $\Rightarrow \underline{Y} \sim N(\underline{0}, \Sigma)$

Results

1. $\underline{X} \sim N(\underline{\mu}, \underline{\Sigma})$

$\underline{X}' \underline{\Sigma}^{-1} \underline{X} \sim \text{Non-Centrality } \chi^2(p, \underline{\mu}' \underline{\Sigma}^{-1} \underline{\mu})$

C : non-singular matrix such that $C \underline{\Sigma}^{-1} C^T = I$

$\underline{Z} = C \underline{X} \sim N(C \underline{\mu}, C \underline{\Sigma}^{-1} C^T)$

$\sim N(\underline{C \mu}, I)$

$\underline{X}' \underline{\Sigma}^{-1} \underline{X} = \underline{Z}' C^{-1} \underline{\Sigma}^{-1} C^{-1} \underline{Z} = \underline{Z}' (C \underline{\Sigma} C^T)^{-1} \underline{Z} = \underline{Z}' \underline{Z}$

Dist of $\underline{X}' \underline{\Sigma}^{-1} \underline{X} = \sum_{i=1}^p Z_i^2 \sim \text{Non-central } \chi^2(p, \underline{\mu}' \underline{\Sigma}^{-1} \underline{\mu})$

$$(\bar{\underline{X}} - \underline{\mu})' \left(\frac{\underline{\Sigma}}{N} \right)^{-1} (\bar{\underline{X}} - \underline{\mu}) \sim \chi^2(r)$$

$$\sqrt{N}(\bar{\underline{X}} - \underline{\mu}) \sim N(\underline{0}, \underline{\Sigma})$$

$$N(\bar{\underline{X}} - \underline{\mu})' \underline{\Sigma}^{-1} (\bar{\underline{X}} - \underline{\mu}) \sim \chi^2(r) \rightarrow \text{This can be used to test } \underline{\mu} = \underline{\mu}_0, \text{ when } \underline{\Sigma} \text{ is known.}$$

Hotelling T^2

THEOREM: If A and B are two real symmetric matrices $\times$

Wishart Distribution

$$\underline{x}_1, \dots, \underline{x}_N \sim N(\underline{\mu}, \underline{\Sigma}), \quad \hat{\underline{\mu}} = \bar{\underline{x}}, \quad \hat{\underline{\Sigma}} = S = \frac{A}{N-1} = \frac{A}{n}$$

$$A = \sum_{\alpha=1}^N (\underline{x}_\alpha - \bar{\underline{x}})(\underline{x}_\alpha - \bar{\underline{x}})^T$$

$$n = N-1 \quad (N-1)A = S$$

$$S = \begin{pmatrix} s_{11} & s_{12} & \dots & s_{1p} \\ \vdots & s_{22} & \dots & s_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ s_{p1} & s_{p2} & \dots & s_{pp} \end{pmatrix}$$

$p(p+1)/2$ elements

Let $\underline{x}_1, \underline{x}_2, \dots, \underline{x}_N$ ($N > p$) are independent of each other, $N(\underline{\mu}, \underline{\Sigma})$; if a p.d. matrix A has the pdf

$$f(A) = |A|^{\frac{p-p-1}{2}} \exp\left(-\frac{1}{2} \text{tr}(\underline{\Sigma}^{-1} A)\right)$$

$$\frac{2^{\frac{np}{2}}}{|\underline{\Sigma}|^{\frac{n}{2}}} \pi^{\frac{p(p-1)}{4}} \prod_{i=0}^{p-1} \Gamma\left(\frac{n-i}{2}\right)$$

then A is said to have a Wishart distribution, with parameters $\underline{\Sigma}$ and n . $W(\underline{\Sigma}, n)$

Theorem: If A and B are two real symmetric matrices such that B is p.d., then there exists a non singular matrix P st. $P'AP = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$, $P'BP = I_n$ where $\lambda_1, \lambda_2, \dots, \lambda_n$ are roots of the eqn $|A - \lambda B| = 0$

Pf. B is a pd, $\exists$ a non singular matrix Q st. $Q'BQ = I$
 Consider $Q'AQ$; real, symmetric, so by spectral decomposition thm., $\exists$ an orthogonal matrix H st. $H'(Q'AQ)H = D = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$ where $\lambda_1, \lambda_2, \dots, \lambda_n$ are roots of $|Q'AQ - \lambda I| = 0$.

Let $P = QH$ (P is n.s.) $\Rightarrow P'AP = D = \text{Diag}(\lambda_1, \dots, \lambda_n)$

Q is n.s., $\Rightarrow P$ is also n.s.

$$Q'PQ = I \quad \text{or} \quad H'Q'BQH = H'H$$

$$\text{or} \quad P'BP = I_n$$

Consider $|Q'AQ - \lambda I|$

$$\text{or} \quad |Q'AQ - \lambda Q'BQ| = 0$$

$$\text{or} \quad |Q'| |A - \lambda I| |Q| = 0$$

$$\text{or} \quad |QQ'| |A - \lambda I| \quad \text{or} \quad |A - \lambda I| = 0.$$

→ # Thm: If $y_\alpha \sim N_p(\underline{0}, \Sigma)$, $\alpha = 1, 2, \dots, n$ and are independently distributed then the density of $A = \sum_{\alpha=1}^n y_\alpha y_\alpha' \sim W(\Sigma, n)$ for pd. A and $n \geq p$
 (For $n < p$ no density fn).

Characteristic fn

Let $y_\alpha \stackrel{\text{ind}}{\sim} N(\underline{0}, \Sigma)$ $\alpha = 1, 2, \dots, n$; $A = \sum_{\alpha=1}^n y_\alpha y_\alpha'$

Ch. fn. $\phi(\Theta) = E[\exp(\text{tr } A \Theta)]$
 symmetric matrix

$$\begin{aligned}
 \phi(\Theta) &= E \left[\exp \left(i \operatorname{tr} \left(\sum_{\alpha=1}^n \underline{y}_{\alpha} \underline{y}_{\alpha}' \Theta \right) \right) \right] \\
 &= E \left[\exp \left(i \sum_{\alpha=1}^n \underline{y}_{\alpha}' \Theta \underline{y}_{\alpha} \right) \right] \\
 &= \prod_{\alpha=1}^n E \left[\exp (i \underline{y}_{\alpha}' \Theta \underline{y}_{\alpha}) \right] \\
 &= \{ E [\exp (i \underline{y}' \Theta \underline{y})] \}^n
 \end{aligned}$$

~ Consider

$$E [\exp (i \underline{y}' \Theta \underline{y})]$$

Consider Σ^{-1} and Θ real, $\exists$ a real n.s. matrix P st. $P' \Sigma^{-1} P = I$, $P' \Theta P = \operatorname{diag} (\lambda_1, \lambda_2, \dots, \lambda_p) = D$

, where $\lambda_1, \lambda_2, \dots, \lambda_p$ are the ch. roots of $|\Theta - \lambda I| = 0$

Now,

$$\underline{Z} \sim N(0, I)$$

Consider a linear transformation

$$\underline{y} = P \underline{Z}, \quad Z_j \text{ are independently dist.}$$

$$E [\exp (i \underline{y}' \Theta \underline{y})] = E \left(\exp (i \underline{Z}' \underbrace{P' \Theta P}_D \underline{Z}) \right)$$

$$= E \left[\exp \left(i \sum_{j=1}^p \lambda_j Z_j^2 \right) \right]$$

$$Z_j \sim N(0, 1)$$

$$Z_j^2 \sim \chi^2(1)$$

$$= \prod_{j=1}^p \phi_{\chi^2_1}(Z_j)$$

$$= \prod_{j=1}^p (1 - 2i\lambda_j)^{-1/2}$$

$$= |I - 2iD|^{-1/2}$$

$$= |P' \Sigma^{-1} P - 2i P' \Theta P|^{-1/2}$$

$$= (|P'| | \Sigma^{-1} - 2i \Theta | |P|)^{-1/2}$$

$$= (|P|^2 | \Sigma^{-1} - 2i \Theta |)^{-1/2} = \frac{| \Sigma^{-1} - 2i \Theta |^{-1/2}}{|P'|^{1/2}}$$

(Consider

$$P' I P = \Sigma^{-1} - 2i \Theta |P|)$$

$$P \cdot T \cdot O$$

$$= \frac{|\Sigma^{-1} - 2i\Theta|^{-1/2}}{|\Sigma|^{1/2}}$$

$$= |I - 2i\Theta\Sigma|^{-1/2}$$

$$\phi(\Theta) = |I - 2i\Theta\Sigma|^{-n/2} \sim \text{fn of } \Sigma, n$$

→

$f(A)$ = pdf of Wishart

$$\phi(\Theta) = \int f(A) \exp(i \text{tr} A \Theta) dA$$

→ Additive property

If $A_j (j=1, \dots, q) \stackrel{\text{ind}}{\sim} (\Sigma, n_j)$
 then $A = \sum_{j=1}^q A_j \sim W(\Sigma, \sum_{j=1}^q n_j)$

⇒ Pf:

$$\phi_{A_j}(\Theta) = |I - 2i\Theta\Sigma|^{-n_j/2}$$

$$\phi_A(\Theta) = \phi_{\sum A_j}(\Theta) = \prod_{j=1}^q \phi_{A_j}(\Theta)$$

$$= |I - 2i\Theta\Sigma|^{-\frac{\sum n_j}{2}}$$

$$= |I - 2i\Theta\Sigma|^{-n_1/2} \times |I - 2i\Theta\Sigma|^{-n_2/2} \dots |I - 2i\Theta\Sigma|^{-n_q/2}$$

By uniqueness thm.

$$A \sim W(\Sigma, \sum_{j=1}^q n_j) \quad (\text{Reproductive property})$$

→ Distribution of linear transformation

If $A \sim W(\Sigma, n)$; $C: p \times p$ n.s. matrix then
 for $B = C'AC \sim W(C'\Sigma C, n)$ where C is
 such that $\underline{z}_\alpha = C'\underline{y}_\alpha$; $A = \sum \underline{y}_\alpha \underline{y}_\alpha'$

$$\underline{y}_\alpha \stackrel{\text{ind}}{\sim} N_p(0, \Sigma) \quad \forall \alpha=1, \dots, n$$

$$\underline{z}_\alpha = C'\underline{y}_\alpha$$

$$B = C'AC = C'(\sum \underline{y}_\alpha \underline{y}_\alpha')C = \sum (C'\underline{y}_\alpha)(C'\underline{y}_\alpha)' = \sum \underline{z}_\alpha \underline{z}_\alpha'$$

$$\underline{y}_\alpha \sim N(\underline{0}, \Sigma) \Rightarrow \underline{z}_\alpha \sim N(\underline{0}, C' \Sigma C)$$

$$\Rightarrow \sum \underline{z}_\alpha \underline{z}_\alpha' \sim W(C' \Sigma C, n)$$

→ Marginal distribution

$$A = \sum \underline{y}_\alpha \underline{y}_\alpha'$$

$$\underline{y}_\alpha \stackrel{\text{ind}}{\sim} N(\underline{0}, \Sigma)$$

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$$

(q × q) (p-q) × (p-q)

$$\begin{pmatrix} \underline{y}_\alpha^{(1)} \\ \underline{y}_\alpha^{(2)} \end{pmatrix} \rightarrow \begin{matrix} q \\ (p-q) \end{matrix}$$

$$\Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}$$

$$A_{11} \sim W_q(\Sigma_{11}, n)$$

$$\rightarrow \underline{y}_\alpha = \begin{pmatrix} \underline{y}_\alpha^{(1)} \\ \underline{y}_\alpha^{(2)} \end{pmatrix} \quad \forall \quad \alpha = 1, 2, \dots, n$$

$$\underline{y}_\alpha^{(1)} \stackrel{\text{ind}}{\sim} N_q(\underline{0}, \Sigma_{11})$$

$$A_{11} = \sum_{\alpha=1}^n \underline{y}_\alpha^{(1)} \underline{y}_\alpha^{(1)'} \sim W_q(\Sigma_{11}, n)$$

$$\rightarrow \underline{y}_\alpha = \begin{pmatrix} \underline{y}_\alpha^{(1)} \\ \vdots \\ \underline{y}_\alpha^{(q)} \end{pmatrix} \rightarrow \begin{matrix} p_1 \\ \vdots \\ p_q \end{matrix}$$

$$A = \begin{pmatrix} A_{11} & A_{12} & \dots & A_{1q} \\ A_{21} & A_{22} & \dots & A_{2q} \\ \vdots & \vdots & \ddots & \vdots \\ A_{q1} & A_{q2} & \dots & A_{qq} \end{pmatrix}$$

$$\Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} & \dots & \Sigma_{1q} \\ \Sigma_{21} & \Sigma_{22} & & \vdots \\ \vdots & \vdots & \ddots & \vdots \\ \Sigma_{q1} & \dots & \dots & \Sigma_{qq} \end{pmatrix}$$

• If $\Sigma_{ij} = 0 \quad \forall \quad i \neq j$

and if $A \sim W(\Sigma, n)$

Then $A_{11}, A_{22}, \dots, A_{qq}$ are independently distributed

$$A_{jj} \sim W(\Sigma_{jj}, n)$$

→ If $\Sigma_{ij} = 0 \Rightarrow \underline{y}_\alpha^{(1)}, \underline{y}_\alpha^{(2)}, \dots, \underline{y}_\alpha^{(q)}, \alpha = 1, 2, \dots, n$ are independent

$A_{11} = \sum \underline{y}_\alpha^{(1)} \underline{y}_\alpha^{(1)'}, \dots, A_{qq} = \sum \underline{y}_\alpha^{(q)} \underline{y}_\alpha^{(q)'}$ are also independent.

$$A_{jj} \sim W_{p_j}(\Sigma_{jj}, n) \quad j = 1, \dots, q$$

$$\bullet A_{22 \cdot 1} = A_{22} - A_{21} A_{11}^{-1} A_{12}$$

$$\underline{y}_\alpha \sim N(\underline{0}, \underline{\Sigma}) = \begin{pmatrix} y_\alpha^{(1)} \\ y_\alpha^{(2)} \end{pmatrix} \rightarrow \begin{matrix} a \\ b \end{matrix}$$

$$A = \sum y_\alpha y_\alpha' = Y'Y \quad Y = \begin{pmatrix} y_1' \\ y_2' \\ \vdots \\ y_n' \end{pmatrix}$$

$$\begin{aligned} A_{22 \cdot 1} &= A_{22} - A_{21} A_{11}^{-1} A_{12} \\ &= \underline{y}_2' \underline{y}_2 - \underline{y}_2' \underline{y}_1 A_{11}^{-1} \underline{y}_1' \underline{y}_2 \\ &= \underline{y}_2' (\underline{I} - \underline{y}_1 A_{11}^{-1} \underline{y}_1') \underline{y}_2 \\ &= \underline{y}_2' P \underline{y}_2 \end{aligned}$$

$$P = \underline{I} - \underline{y}_1 A_{11}^{-1} \underline{y}_1' \text{ symmetric, idempotent, rank}(P) = n - q$$

$$\underline{y}_1 P = 0$$

$$\underline{y}_{2 \cdot 1} = \underline{y}_2 - \underline{y}_1 \underline{\Sigma}_{11}^{-1} \underline{\Sigma}_{12}$$

$$\text{Cov}(\underline{y}_{2 \cdot 1}, \underline{y}_1) = 0$$

$$\underline{\Sigma}_{12} - \underline{\Sigma}_{11} \underline{\Sigma}_{11}^{-1} \underline{\Sigma}_{12}$$

$$\underline{y}_{2 \cdot 1}' P \underline{y}_{2 \cdot 1}$$

$$= (\underline{y}_2 - \underline{y}_1 \underline{\Sigma}_{11}^{-1} \underline{\Sigma}_{12})' (\underline{I} - \underline{y}_1 A_{11}^{-1} \underline{y}_1') (\underline{y}_2 - \underline{y}_1 \underline{\Sigma}_{11}^{-1} \underline{\Sigma}_{12})$$

$$\Rightarrow \underline{y}_2' \underline{y}_2 - \underline{y}_2' \underline{y}_1 A_{11}^{-1} \underline{y}_1' \underline{y}_2 = \underline{y}_2' P \underline{y}_2$$

$$= A_{22 \cdot 1}$$

$$\underline{y}_{2 \cdot 1} \sim N(\mu_{2 \cdot 1}, \underline{\Sigma}_{11 \cdot 2})$$

$$\sum_{\alpha=1}^n \underline{y}_{2 \cdot 1} \underline{y}_{2 \cdot 1}' = A_{11 \cdot 2} \sim W(\underline{\Sigma}_{11 \cdot 2}, n)$$

Generalized varianceMultivariate analogue of σ^2 is $|\Sigma|$ Another " " σ^2 " $|\Sigma|$ Sample version of $|\Sigma|$ is $|\hat{S}| = \left| \frac{1}{(N-1)^{p+1}} \sum_{\alpha=1}^N (\underline{x}_\alpha - \bar{\underline{x}})(\underline{x}_\alpha - \bar{\underline{x}})' \right|$

generalized variance.

→ Given a random sample $\underline{x}_1, \underline{x}_2, \dots, \underline{x}_N$ from $N_p(\underline{\mu}, \Sigma)$, then dist of $\frac{|\hat{S}|}{(N-1)^p}$ is the same as the dist.

of $\frac{|\Sigma|}{(N-1)^p}$ times the product of independent factors where i^{th} factor has $\chi^2(N-i)$ dist.

$$f(|\hat{S}|) \sim \frac{|\Sigma|}{n^p} f(\chi_1^2) \cdot f(\chi_2^2) \cdots f(\chi_{n-p+1}^2)$$

Asymptotic dist

$$\sqrt{n} \left(\frac{|\hat{S}|}{|\Sigma|} - 1 \right) \xrightarrow{\text{Asy}} N(0, 2p)$$

$$E(|\hat{S}|) = \frac{|\Sigma|}{n^p} \pi E(\chi_{N-i}^2) = \frac{|\Sigma|}{n^p} \prod_{i=1}^p (N-i)$$

Cochran's TheoremSuppose $\underline{y}_1, \underline{y}_2, \dots, \underline{y}_N \stackrel{\text{i.i.d.}}{\sim} N(\underline{0}, \Sigma)$ $\sum_{i=1}^N C_i = I_N$ Suppose C_i : $N \times N$ symmetric matrix, $\text{Rank}(C_i) = r_i$ Suppose $C_i = (C_{i\alpha\beta})$, $Q_i = \sum_{\alpha=1}^N \sum_{\beta=1}^N C_{i\alpha\beta} \underline{y}_\alpha \underline{y}_\beta'$

$$\sum_{i=1}^m Q_i = \sum_{\alpha=1}^N \underline{y}_\alpha \underline{y}_\alpha'$$

A necessary and sufficient condition that $Q_1, \dots, Q_m$ are independently distributed with $Q_i \sim W(\Sigma, r_i)$ is $\sum_{i=1}^m r_i = N$

Inverted Wishart distribution

$$A \sim W(\Sigma, n)$$

$$B = A^{-1} \sim f(B) = \frac{|\Sigma^{-1}|^{n/2} |B|^{-(n+p+1)/2}}{2^{np/2} \pi^{p(p-1)/4} \prod_{i=0}^{p-1} \Gamma(n-i)} \exp\left(-\frac{1}{2} \text{tr} \Sigma^{-1} B^{-1}\right)$$

Non-central Wishart

$$\underline{y}_\alpha \stackrel{\text{ind}}{\sim} N(\underline{0}, \Sigma)$$

$$\sum \underline{y}_\alpha \underline{y}_\alpha' \sim W(\Sigma, n)$$

$$\underline{x}_\alpha \stackrel{\text{ind}}{\sim} N(\underline{M}, \Sigma)$$

$$\underline{A} = \sum_{\alpha=1}^n \underline{x}_\alpha \underline{x}_\alpha' \sim \text{Non-central Wishart}(\Sigma, n, \Sigma^{-1} \underline{M} \underline{M}')$$

Result:

$$A \sim W(\Sigma, n) ; E(A) = n \Sigma \quad E(A^{-1}) = \frac{\Sigma^{-1}}{n-p-1}$$

Generalized T^2 statistics

Univariate case:

$$X_1, X_2, \dots, X_N \sim N(\mu, \sigma^2)$$

$$t = \frac{\sqrt{N}(\bar{X} - \mu)}{s} \sim t(N-1)$$

Multivariate analogue of t

$$T^2 = N(\bar{X} - \underline{\mu})' S^{-1} (\bar{X} - \underline{\mu})$$

Hotelling T^2 statistic, (or Generalized T^2 statistic)
used for test of hypothesis.

Likelihood ratio test

$$N(\underline{\mu}, \underline{\Sigma}) ; H_0 : \underline{\mu} = \underline{\mu}_0$$

$$\lambda = \frac{\text{Max}_{H_0} L}{\text{Max}_{\Omega} L} = \frac{\text{Max} L(\underline{\mu}_0, \underline{\Sigma})}{\text{Max} L(\underline{\mu}, \underline{\Sigma})}$$

Parametric space under $H_0 = \hat{\Omega}_{H_0} = (\underline{\mu}_0, \underline{\Sigma}_{00})$

whole space under $H_1 = \hat{\Omega}_{H_1} = (\hat{\underline{\mu}}_n, \hat{\underline{\Sigma}}_n)$

$$\hat{\underline{\mu}}_n = \bar{\underline{x}} \quad \hat{\underline{\Sigma}}_n = \frac{A}{N} \quad \hat{\underline{\Sigma}}_{00} = \frac{1}{N} \sum_{\alpha=1}^N (\underline{x}_{\alpha} - \underline{\mu}_0)(\underline{x}_{\alpha} - \underline{\mu}_0)'$$